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Lecture 04 - Step 1

Linear Regression (Part 1)

Year 02
Semester 01
Date : .....

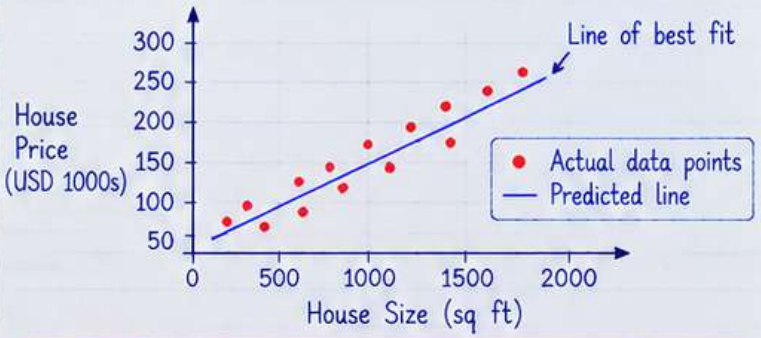
"Small Steps Big Progress" ☺

1 What is Linear Regression?

- Linear regression is a supervised learning algorithm used to predict a continuous numerical value.
- It finds a linear relationship between the input feature(s) (independent variable) and the output (dependent variable).
- The model tries to find the best fitting line that minimizes the error between actual and predicted values.
- It is simple but very powerful and widely used in real world applications (e.g. price prediction, sales forecasting, etc.)

2 Example (Real World)

Predict house price based on house size.



As the house size increases, the house price also increases (positive linear relationship).

3 Types of Linear Regression

1 Simple Linear Regression

- Uses one independent variable (one feature).
- Model equation:

$$y = b_0 + b_1 x$$

where,

- y - predicted value (dependent variable)
- x - input feature (independent variable)
- b_0 - intercept (value of $\hat{y}$ when $x = 0$)
- b_1 - slope (change in $\hat{y}$ for one unit change in x)

2 Multiple Linear Regression

- Uses more than one independent variable.
- Model equation:

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

where,

- $x_1, x_2, \ldots, x_n$ - input features
- b_0 - intercept
- $b_1, b_2, \ldots, b_n$ - coefficients (weights)

When to use Linear Regression?

- ✅ When the target variable is continuous (numeric).
- ✅ When there is a linear relationship between input and output.
- ✅ It is easy to interpret and fast to train.
- ✅ Widely used in real world problems.

4 Best Fit Line (Least Squares Method)

- The best fit line is the line that minimizes the sum of squared errors (SSE) between actual and predicted values.
- This method is called Ordinary Least Squares (OLS).
- For simple linear regression, the coefficients are:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \qquad b_0 = \bar{y} - b_1 \bar{x}$$

where,

- $\bar{x}$ - mean of x values
- $\bar{y}$ - mean of y values

5 Model Assumptions

- Linear relationship between input and output.
- Independence of observations.
- Homoscedasticity (constant variance of errors).
- Normally distributed errors.
- No or little multicollinearity (for multiple regression).

6 Advantages and Disadvantages

Advantages	Disadvantages
✅ Simple and easy to understand.	❌ Only works well for linear relationships.
✅ Fast to train.	❌ Sensitive to outliers.
✅ Works well when the relationship is linear.	❌ Not suitable for complex non-linear data.
✅ Easy to interpret the results.	❌ Assumes certain conditions (e.g. normality, homoscedasticity).

"A simple model can still give powerful results." ☺

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Lecture 04 – Step 1B

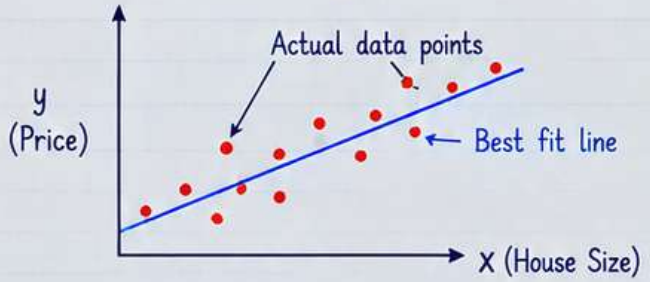
Linear Regression (Part 2)

Best Fit Line and Ordinary Least Squares (OLS)

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Semester 01
Date : .....

1 What is the Best Fit Line?

- The best fit line is the line that best represents the relationship between the input and output data.
- It minimizes the sum of squared errors (SSE) between the actual values and the predicted values.
- This method is called Ordinary Least Squares (OLS).
- The line is used to make predictions for new data.



2 Ordinary Least Squares (OLS)

- OLS finds the values of b_0 and b_1 that minimize the Sum of Squared Errors (SSE).
- SSE is the sum of the squared differences between the actual values and the predicted values.

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - (b_0 + b_1 x_i))^2$$

where,

- y_i – actual value
- $\hat{y}_i$ – predicted value
- x_i – input value
- b_0 – intercept
- b_1 – slope

Goal of OLS:
Find b_0 and b_1 that make the SSE as small as possible (minimum).

3 Formulas for the Coefficients

- For simple linear regression, the coefficients can be calculated using the following formulas:

$$b_1 = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{n\sum x_i^2 - (\sum x_i)^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

where,

- n – number of data points
- $\bar{x}$ – mean of x values
- $\bar{y}$ – mean of y values

4 Example Calculation

- A real estate agent collected the following data:

House Size, x (sq ft)	800	1000	1200	1400	1600
Price, y (USD 1000s)	80	110	150	180	200

Step 1: Calculate the required sums.

$n = 5$
$\sum x = 6000$
$\sum y = 720$
$\sum x^2 = 7{,}600{,}000$
$\sum xy = 912{,}000$

Step 2: Calculate b_1.

$$b_1 = \frac{5(912{,}000) - (6000)(720)}{5(7{,}600{,}000) - (6000)^2}$$

$$b_1 = \frac{4{,}560{,}000 - 4{,}320{,}000}{38{,}000{,}000 - 36{,}000{,}000}$$

$$b_1 = \frac{240{,}000}{2{,}000{,}000} = 0.12$$

Step 3: Calculate b_0.

$$\bar{x} = \frac{6000}{5} = 1200 \qquad \bar{y} = \frac{720}{5} = 144$$

$$b_0 = 144 - 0.12(1200)$$

$$b_0 = 144 - 144 = 0$$

Final model:

$$\hat{y} = 0 + 0.12x$$

$$\hat{y} = 0.12x$$

5 Interpretation of the Coefficients

- b_1 (slope) = 0.12
  → For each additional 1 sq ft increase in house size, the price increases by 0.12 thousand USD (= 120 USD).
- b_0 (intercept) = 0
  → When the house size is 0, the predicted price is 0 (in this example).

6 Making a Prediction

Predict the price for a house with 1800 sq ft.

$$\hat{y} = 0.12(1800) = 216$$

So, the predicted price is 216 thousand USD (= 216,000 USD).

7 Key Points to Remember

- The best fit line is found using Ordinary Least Squares (OLS).
- OLS minimizes the sum of squared errors (SSE).
- Use the formulas to calculate b_1 and b_0.
- A positive slope means a positive linear relationship.
- The model can be used to predict values for new data.
- Simple linear regression uses one input feature.
- Always interpret the coefficients in the context of the problem.

Remember!
The best fit line minimizes the sum of squared errors (SSE).

"The best fit line is the simplest model that explains the data."

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Lecture 04 - Step 1C

Linear Regression (Part 3)

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Semester 01
Date : .....

Calculations - Find the Regression Equation and Make Predictions

1 Given Data

A real estate agent collected data on house size and price.

House Size, x (sq ft)	Price, y (1000 USD)
800	80
1000	110
1200	150
1400	180
1600	200

We will find:
- b_1 (slope)
- b_0 (intercept)
- regression equation
- predict price for a new house

2 Calculate the Required Sums

x	y	x^2	y^2	xy
800	80	640,000	6,400	64,000
1000	110	1,000,000	12,100	110,000
1200	150	1,440,000	22,500	180,000
1400	180	1,960,000	32,400	252,000
1600	200	2,560,000	40,000	320,000
Σ 6000	720	7,600,000	113,400	926,000

$n = 5$ $\quad \Sigma x = 6000$ $\quad \Sigma y = 720$
$\Sigma x^2 = 7{,}600{,}000$ $\quad \Sigma y^2 = 113{,}400$ $\quad \Sigma xy = 926{,}000$

3 Calculate the Slope (b_1)

Use the formula:

$$b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Substitute the values:

$$b_1 = \frac{5(926{,}000) - (6000)(720)}{5(7{,}600{,}000) - (6000)^2}$$
$$= \frac{4{,}630{,}000 - 4{,}320{,}000}{38{,}000{,}000 - 36{,}000{,}000}$$
$$= \frac{310{,}000}{2{,}000{,}000}$$
$$b_1 = 0.155$$

The slope (b_1) tells us how much the price increases for each 1 sq ft increase in house size.

4 Calculate the Intercept (b_0)

Use the formula:

$$b_0 = \bar{y} - b_1\bar{x}$$

$$\bar{x} = \frac{\sum x}{n} \qquad \bar{y} = \frac{\sum y}{n}$$

First find $\bar{x}$ and $\bar{y}$:

$$\bar{x} = \frac{6000}{5} = 1200 \qquad \bar{y} = \frac{720}{5} = 144$$

Now calculate b_0:

$$b_0 = 144 - (0.155)(1200)$$
$$= 144 - 186$$
$$b_0 = -42$$

The intercept (b_0) is the predicted price when the house size is 0 (in this example).

5 Regression Equation

The simple linear regression equation is:

$$\hat{y} = -42 + 0.155x$$

where,
$\hat{y}$ - predicted house price (1000 USD)
x - house size (sq ft)

6 Make a Prediction

Predict the price of a house with 1800 sq ft.
Substitute $x = 1800$ into the equation:

$$\hat{y} = -42 + 0.155(1800)$$
$$= -42 + 279$$
$$\hat{y} = 237$$

So, the predicted price is 237 thousand USD (= 237,000 USD).

As the house size increases, the price also increases because the slope is positive.

7 Key Points to Remember

- Use the formulas to calculate b_1 and b_0.
- The regression equation can be used to predict values for new data.
- The slope (b_1) shows the change in the output for a one unit increase in the input.
- A positive slope means a positive linear relationship.
- Always check the units when interpreting the results.
- This method is called Ordinary Least Squares (OLS) because it minimizes the sum of squared errors (SSE).

Steps Summary:

1. Calculate the required sums.
2. Find the slope (b_1).
3. Find the intercept (b_0).
4. Write the regression equation.
5. Use the equation to make predictions.
6. Interpret the results.

"Practice calculations to build confidence!"

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Lecture 04 – Step 1
Linear Regression

Year 02
Semester 01

1 What is Linear Regression?

- Linear regression is a supervised learning algorithm used to predict a continuous numerical value.
- It finds a linear relationship between the input feature(s) (independent variable) and the output (dependent variable).
- The model tries to find the best fitting line that minimizes the error between predicted and actual values.
- It is simple but very powerful and widely used in real-world applications.

Example:
Predict house price based on size (area).
Input (X) : house size (sq ft)
Output (Y) : house price (USD)

2 Simple Linear Regression

- Uses one independent variable.
- Model equation:

$$y = b_0 + b_1 x$$

where,
y – predicted value (dependent variable)
x – input feature (independent variable)
b_0 – intercept (value of y when $x = 0$)
b_1 – slope (change in y for one unit change in x)

If $b_1 > 0$ → positive relationship
If $b_1 < 0$ → negative relationship

3 Multiple Linear Regression

- Uses more than one independent variable.
- Model equation:

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

where,
$x_1, x_2, \ldots, x_n$ – input features
b_0 – intercept
$b_1, b_2, \ldots, b_n$ – coefficients (weights)

Example:
Predict house price using size, number of bedrooms, and age.

$$\text{Price} = b_0 + b_1(\text{Size}) + b_2(\text{Bedrooms}) + b_3(\text{Age})$$

4 Best Fit Line (Least Squares Method)

- The best fit line is the line that minimizes the sum of squared errors (SSE) between actual and predicted values.
- This method is called Ordinary Least Squares (OLS).
- Formulas for simple linear regression:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \qquad b_0 = \bar{y} - b_1 \bar{x}$$

where,
$\bar{x}$ – mean of x values
$\bar{y}$ – mean of y values

5 Example Calculation (Simple Linear)

- Suppose we have the following data (Hours studied vs Marks):

x (Hours)	1	2	3	4	5
y (Marks)	50	55	65	70	80

Step 1: Find means $\bar{x} = 3$, $\bar{y} = 64$

Step 2: Calculate b_1

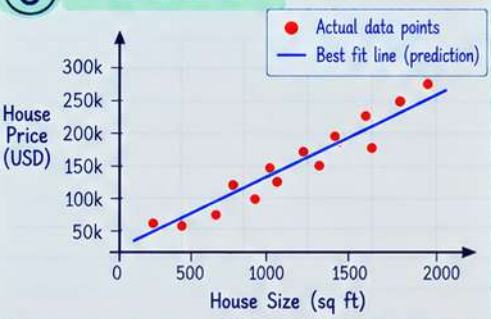
$$b_1 = \frac{\sum (x_i - 3)(y_i - 64)}{\sum (x_i - 3)^2} = \frac{75}{10} = 7.5$$

Step 3: Calculate b_0

$$b_0 = 64 - 7.5(3) = 41.5$$

So the regression line is: $\hat{y} = 41.5 + 7.5x$

6 Visualization



The model finds the line that best fits the data by minimizing the errors.

7 Model Evaluation (Regression)

- Common evaluation metrics:

Metric	Formula	Meaning
Mean Absolute Error (MAE)	$MAE = \frac{1}{n}\sum \lvert y_i - \hat{y}_i \rvert$	Average of absolute errors
Mean Squared Error (MSE)	$MSE = \frac{1}{n}\sum (y_i - \hat{y}_i)^2$	Average of squared errors
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{MSE}$	Square root of MSE (same units as target)
R^2 Score	$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$	Proportion of variance explained (closer to 1 is better)

8 Key Points to Remember

1. Linear regression predicts a continuous value.
2. Simple linear regression uses one feature.
3. Multiple linear regression uses multiple features.
4. Equation: $y = b_0 + b_1 x$ (or with more features).
5. Coefficients are found using the least squares method (OLS).
6. Common metrics: MAE, MSE, RMSE, R^2.
7. R^2 close to 1 means a better fit.
8. Assumes a linear relationship between input and output.
9. Sensitive to outliers.
10. Widely used in real-world problems (e.g. price prediction, sales forecast, etc.).

When to use Linear Regression?

- ☑ When the target variable is continuous (numeric).
- ☑ When there is a linear relationship between input and output.
- ☑ It is simple, fast and easy to interpret.

"Find the simplest model that explains the data well."

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Lecture 04 - Step 2A

Logistic Regression (Part 1)

Introduction, Binary Classification and Sigmoid Function

Year 02
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"Small Steps Big Progress" ☺

1 What is Logistic Regression?

- Logistic regression is a supervised learning algorithm used for classification problems.
- It is mainly used to predict a categorical output (usually binary: 0 or 1).
- Instead of predicting a continuous value like linear regression, it predicts the probability that an input belongs to a particular class.
- It uses the logistic (sigmoid) function to convert the output into a probability between 0 and 1.

Example:
Predict whether a student will pass (1) or fail (0) based on study hours.

Input (X) : Study hours
Output (Y) : Pass (1) or Fail (0)

2 Binary Classification

- Logistic regression is commonly used for binary classification problems (two classes).
- The output is a probability between 0 and 1.
- We use a threshold (usually 0.5) to convert the probability into a class label.

If probability $\geq 0.5 \rightarrow$ class 1
If probability $< 0.5 \rightarrow$ class 0

Example (Pass or Fail):
If predicted probability = 0.72 (≥ 0.5) → Predict Pass (1)
If predicted probability = 0.30 (< 0.5) → Predict Fail (0)

3 Logistic Regression Model

- The model equation is:

$$z = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

- The probability is calculated using the sigmoid function:

$$p = \frac{1}{1 + e^{-z}}$$

where,

p – predicted probability (between 0 and 1)
z – linear combination of inputs
b_0 – intercept
$b_1, b_2, \ldots, b_n$ – coefficients (weights)
$x_1, x_2, \ldots, x_n$ – input features
e – Euler's number (≈ 2.718)

4 Sigmoid Function

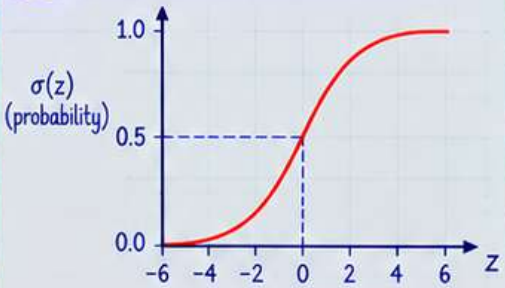
- The sigmoid function converts any real value into a probability between 0 and 1.
- Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Properties:

- Output is always between 0 and 1.
- As z increases, $\sigma(z)$ approaches 1.
- As z decreases, $\sigma(z)$ approaches 0.
- It has an S-shaped curve.

5 Graph of the Sigmoid Function



When $z = 0$, $\sigma(z) = 0.5$
When z is large positive, $\sigma(z) \approx 1$
When z is large negative, $\sigma(z) \approx 0$

6 Example Calculation

Suppose we have a simple model:

$$z = -3 + 0.8x$$

For a student who studies 5 hours ($x = 5$):

Step 1: Calculate z

$z = -3 + 0.8(5) = -3 + 4 = 1$

Step 2: Calculate probability

$$p = \frac{1}{1 + e^{-1}} = \frac{1}{1 + 0.368} = 0.731$$

Step 3: Determine class

Since $0.731 \geq 0.5 \rightarrow$ Predict Pass (1)

7 Comparision with Linear Regression

Feature	Linear Regression	Logistic Regression
Type	Regression	Classification
Output	Continuous value	Probability (0 to 1) → class (0 or 1)
Model	$y = b_0 + b_1 x$	$p = \frac{1}{1 + e^{-z}}$
Output range	$(-\infty, +\infty)$	0 to 1
Used for	Predicting numerical values	Predicting categorical classes
Example	House price (USD)	Pass or Fail (0 or 1)

8 Key Points to Remember

1. Logistic regression is used for classification (mainly binary).
2. It predicts the probability of a class.
3. The sigmoid function converts the output to a value between 0 and 1.
4. Use a threshold (usually 0.5) to determine the class.
5. The model equation is $z = b_0 + b_1 x_1 + \ldots + b_n x_n$
6. $p = \frac{1}{1 + e^{-z}}$
7. If $p \geq 0.5 \rightarrow$ class 1, if $p < 0.5 \rightarrow$ class 0.
8. Widely used in real-world problems (e.g. medical diagnosis, spam detection, customer churn, etc.).

When to use Logistic Regression?

- ☑ When the output is categorical (0 or 1).
- ☑ When you need to predict the probability of a class.
- ☑ When there is a linear relationship between input and the log-odds of the output.
- ☑ Widely used in real-world classification problems.

☺

"Logistic regression doesn't predict the class directly, it predicts the probability."

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Lecture 04 - Step 2D

Logistic Regression (Part 4)

Summary, Evaluation, Example Problems and Exam Points

Year 02
Semester 01
Date : .....

"Practice Makes Perfect" ☺

1 Complete Prediction Process (Summary)

Step 1 Take input features $x_1, x_2, \ldots, x_n$

Step 2 Calculate the model score (z)

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

Step 3 Convert to probability using sigmoid function

$$p = \frac{1}{1 + e^{-z}}$$

Step 4 Get the probability $(0 \le p \le 1)$

Step 5 Compare with cutoff value (usually 0.5)

Step 6 Assign class:

If $p \ge 0.5$ → class 1
If $p < 0.5$ → class 0

2 Example Question 1 (Single Feature)

A model is given by $z = -4 + 0.8x$
Find the probability and predicted class for a student who studies 6 hours. Use a cutoff value of 0.5.

Step 1: Calculate z

$z = -4 + 0.8(6)$
$z = -4 + 4.8$
$z = 0.8$

Step 2: Calculate probability

$$p = \frac{1}{1 + e^{-0.8}} = \frac{1}{1 + 0.449} = \frac{1}{1.449} = 0.690$$

Step 3: Predict class

Since $0.690 \ge 0.5$ → class 1 (Pass)

The student is predicted to pass with a probability of 0.690.

3 Example Question 2 (Multiple Features)

A model is given by $z = -2 + 0.5x_1 + 1.2x_2$
For a person with $x_1 = 3$, $x_2 = 1$,
find the probability and predicted class.

Step 1: Calculate z

$z = -2 + 0.5(3) + 1.2(1)$
$z = -2 + 1.5 + 1.2 = 0.7$

Step 2: Calculate probability

$$p = \frac{1}{1 + e^{-0.7}} = \frac{1}{1 + 0.497} = \frac{1}{1.497} = 0.668$$

Step 3: Predict class

$0.668 \ge 0.5$ → class 1

The person is predicted to belong to class 1 with probability 0.668.

4 Evaluation Metrics for Classification

- For classification problems, we use a confusion matrix to evaluate the model.
- From the confusion matrix, we can calculate:
  - Accuracy
  - Precision
  - Recall
  - F1-Score

Confusion Matrix	Predicted 0	Predicted 1
Actual 0	TN (True Negative)	FP (False Positive)
Actual 1	FN (False Negative)	TP (True Positive)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

5 Example: Confusion Matrix and Metrics

- Suppose a model gives the following confusion matrix on a test set:

	Predicted 0	Predicted 1
Actual 0	50	10
Actual 1	5	35

$$\text{Accuracy} = \frac{50 + 35}{50 + 10 + 5 + 35} = \frac{85}{100} = 0.85$$

$$\text{Precision} = \frac{35}{35 + 10} = \frac{35}{45} = 0.778$$

$$\text{Recall} = \frac{35}{35 + 5} = \frac{35}{40} = 0.875$$

$$\text{F1-Score} = \frac{2(0.778)(0.875)}{0.778 + 0.875} = 0.824$$

The model has 85% accuracy on the test set.

6 Logistic Regression vs Linear Regression

Aspect	Linear Regression	Logistic Regression
Type of problem	Regression (numerical prediction)	Classification (categorical prediction)
Output	Continuous value	Probability (0 to 1)
Model equation	$y = \beta_0 + \beta_1 x + \ldots$	$p = \frac{1}{1 + e^{-z}}$
Estimation method	Ordinary Least Squares (OLS)	Maximum Likelihood Estimation (MLE)
Output range	$(-\infty, +\infty)$	0 to 1
Decision	Not required	Use cutoff (e.g. 0.5)
Example	Predict house price	Predict pass or fail

7 Key Points to Remember

- ☑ Logistic regression is used for classification problems.
- ☑ It predicts the probability of a class.
- ☑ The sigmoid function converts the output to a value between 0 and 1.
- ☑ A cutoff value (usually 0.5) is used to determine the class.
- ☑ Model parameters are estimated using Maximum Likelihood Estimation (MLE).
- ☑ Evaluate the model using a confusion matrix and metrics (accuracy, precision, recall, F1).
- ☑ Choose appropriate evaluation metrics based on the problem (e.g, use recall for imbalanced data).

8 Common Exam Questions

1. Explain what logistic regression is and how it works.
2. Write and explain the sigmoid function.
3. What is the logit function?
4. How are the model parameters estimated?
5. Given a model and input values, calculate the probability and predicted class.
6. Given a confusion matrix, calculate evaluation metrics.
7. Compare linear regression and logistic regression.
8. Discuss situations where logistic regression is useful.

"Understand the concepts, practice the calculations, and you'll do well!" ☺

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Lecture 04 – Step 2D

Logistic Regression (Part 4)

Example Calculation, Evaluation, Confusion Matrix and Key Points

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Semester 01
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"Practice Makes Progress" ☺

1 Example Calculation

A model is given as:

$$z = -3 + 0.8x$$

Suppose a student studies 5 hours ($x = 5$).

Step 1: Calculate z

$$z = -3 + 0.8(5) = -3 + 4 = 1$$

Step 2: Calculate probability using sigmoid

$$p = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-1}} = \frac{1}{1+0.368} = 0.731$$

Step 3: Determine class (cutoff = 0.5)

Since $0.731 \geq 0.5 \rightarrow$ Predict Class 1

So, a student who studies 5 hours is predicted to Pass (1).

2 Another Example

Use the same model: $z = -3 + 0.8x$

For a student who studies 2 hours ($x = 2$):

$$z = -3 + 0.8(2) = -3 + 1.6 = -1.4$$

$$p = \frac{1}{1+e^{-(-1.4)}} = \frac{1}{1+e^{1.4}}$$

$$= \frac{1}{1+4.055} = 0.198$$

Since $0.198 < 0.5 \rightarrow$ Predict Class 0

So, a student who studies 2 hours is predicted to Fail (0).

Key Idea:

- Higher study hours → higher probability.
- Lower study hours → lower probability.
- The sigmoid function always gives a value between 0 and 1.

3 Model Interpretation

The model is: $z = -3 + 0.8x$

- Intercept ($\beta_0 = -3$)
  - When study hours = 0, the log-odds is −3.
  - This means the initial probability of passing is low.
- Coefficient ($\beta_1 = 0.8$)
  - For each additional 1 hour of study, the log-odds of passing increases by 0.8.
  - A positive coefficient means a positive relationship.

In logistic regression, coefficients represent the change in log-odds (not the direct change in probability).

4 Confusion Matrix

A confusion matrix shows how well the model predicted the classes.

		Actual Class	
		Class 1 (Pass)	Class 0 (Fail)
Predicted Class	Class 1 (Pass)	TP (True Positive)	FP (False Positive)
	Class 0 (Fail)	FN (False Negative)	TN (True Negative)

- TP – correctly predicted as 1
- TN – correctly predicted as 0
- FP – predicted 1 but actually 0
- FN – predicted 0 but actually 1

5 Evaluation Metrics (Classification)

- Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Proportion of correct predictions.

- Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Of the predicted positives, how many are actually positive?

- Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$

Of the actual positives, how many are correctly predicted?

- F1-Score

$$F1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Harmonic mean of precision and recall.

6 Example: Calculate Metrics

Suppose we have the following confusion matrix for a student pass/fail model:

		Actual Class	
		Pass (1)	Fail (0)
Predicted Class	Pass (1)	40	10
	Fail (0)	5	45

Calculate the evaluation metrics:

- Accuracy $= \frac{40 + 45}{40 + 45 + 10 + 5} = \frac{85}{100} = 0.85$
- Precision $= \frac{40}{40 + 10} = \frac{40}{50} = 0.80$
- Recall $= \frac{40}{40 + 5} = \frac{40}{45} = 0.889$
- F1 $= \frac{2 \times (0.80)(0.889)}{0.80 + 0.889} = 0.842$

These metrics help us understand how well the model is performing.

7 Advantages and Disadvantages

Advantages	Disadvantages
✓ Simple and easy to understand.	✗ Only works well for linear relationships (in log-odds).
✓ Works well for binary classification.	✗ Assumes independent features.
✓ Provides probability output.	✗ Not suitable for complex non-linear data.
✓ Less prone to overfitting compared to complex models.	✗ Sensitive to outliers.
✓ Widely used in real-world applications (e.g. spam detection, medical diagnosis).	✗ Requires larger datasets for better performance.

8 Key Points to Remember

1. Logistic regression is used for binary classification.
2. It predicts the probability of a class (0 or 1).
3. The sigmoid function converts the output to a value between 0 and 1.
4. Common cutoff value is 0.5 (can be changed).
5. Model is estimated using Maximum Likelihood Estimation (MLE).
6. Evaluate using confusion matrix and metrics (accuracy, precision, recall, F1-score).
7. Coefficients represent change in log-odds.
8. Widely used in real-world classification problems.

"Logistic regression turns numbers into probabilities, and probabilities into decisions." ☺

Keep Learning !

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Lecture 04 - Step 3A

K-Nearest Neighbors (KNN) (Part 1)

Introduction, How it Works and Distance Calculation

Year 02
Semester 01
Date : .....

"Simple Idea
Powerful Algorithm" ☺

1 What is K-Nearest Neighbors (KNN)?

- KNN is a supervised learning algorithm used for classification (and also regression).
- It predicts the class of a new data point based on the majority class of its K nearest neighbors.
- It is a simple but powerful algorithm.
- It is called "instance-based learning" because it does not build an explicit model. It just stores the training data.

Main idea: "Similar data points are likely to belong to the same class."

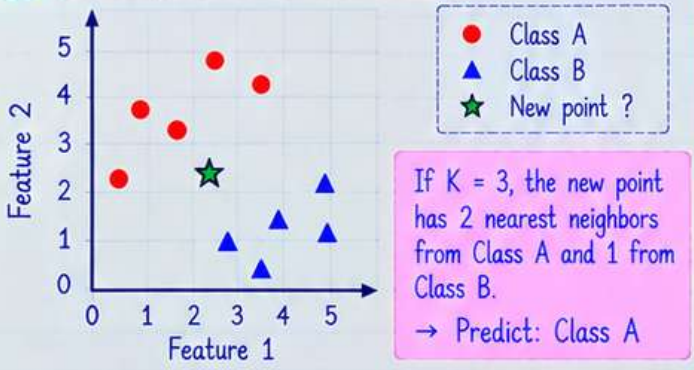
2 How KNN Works?

1. Choose the value of K (number of neighbors).
2. Calculate the distance between the new data point and all training data points.
3. Find the K nearest neighbors (smallest distances).
4. Look at the classes of these neighbors.
5. For classification: predict the class with the majority vote.
6. For regression: predict the average value of the neighbors.

KNN is non-parametric → it does not assume any specific data distribution.

K is usually a small odd number (e.g. 1, 3, 5) to avoid ties in classification.

3 Example (Classification)



If K = 3, the new point has 2 nearest neighbors from Class A and 1 from Class B.
→ Predict: Class A

4 Distance Calculation

- The most common distance measure is the Euclidean distance.
- It measures how far apart two points are in space.

Euclidean Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For n features:

$$d = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

where,
(x_1, y_1) – point 1
(x_2, y_2) – point 2
d – distance between the points

Smaller distance means the points are more similar.

5 Example: Calculate Euclidean Distance

Find the distance between point A(2, 3) and B(5, 7).

$$d = \sqrt{(5-2)^2 + (7-3)^2}$$
$$= \sqrt{(3)^2 + (4)^2}$$
$$= \sqrt{9 + 16}$$
$$= \sqrt{25}$$
$$= 5$$

The distance between A and B is 5.

6 Example with Multiple Features

Find the distance between A(1, 2, 3) and B(4, 6, 8).

$$d = \sqrt{(4-1)^2 + (6-2)^2 + (8-3)^2}$$
$$= \sqrt{(3)^2 + (4)^2 + (5)^2}$$
$$= \sqrt{9 + 16 + 25}$$
$$= \sqrt{50}$$
$$= 7.07 \text{ (approximately)}$$

The distance between A and B is approximately 7.07.

7 Choosing the Value of K

- K is a hyperparameter (chosen by the user).
- A small K (e.g. 1) can be sensitive to noise (outliers).
- A large K can make the decision boundary too smooth and may ignore local patterns.
- Usually an odd number is chosen for classification (e.g. 3, 5, 7).
- The best K can be found using techniques like cross-validation.

There is no single best value for K → it depends on the data!

8 Key Points to Remember

1. KNN is used for classification and regression.
2. It is an instance-based learning algorithm.
3. It predicts based on the K nearest neighbors.
4. Euclidean distance is the most common distance measure.
5. Smaller distance means more similar points.
6. K is a user-defined parameter (usually odd for classification).
7. It is simple to understand but can be computationally expensive for large datasets.
8. It works well when similar data points belong to the same class.

"KNN keeps it simple – learn from your neighbors!" ☺

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Lecture 04 – Step 3B

K-Nearest Neighbors (KNN) (Part 2)

Full Prediction Process with Examples (Classification and Regression)

Year 02
Semester 01
Date : .....

"Learn by Doing Examples" ☺

1 KNN Prediction Process (Steps)

1. Choose the value of K.
2. Calculate the distance between the new data point and all training data points.
3. Sort the distances in ascending order.
4. Select the K nearest neighbors.
5. Use the neighbors to make the prediction.

KNN is a lazy learning algorithm because it does not build a model, it just stores the data and makes predictions when needed.

2 Example Dataset (Classification)

Suppose we have the following training data about students:

ID	Study Hours (x)	Previous Score (y)	Result (Class)
A	1	50	Fail (0)
B	2	55	Fail (0)
C	3	70	Pass (1)
D	4	80	Pass (1)
E	5	90	Pass (1)

We want to predict the class for a new student with study hours = 3 and previous score = 65 using K = 3.

3 Step 1: Calculate Distances

Use Euclidean distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Calculate the distance between the new point (3, 65) and each training point:

ID	(x, y)	Calculation	Distance
A	(1, 50)	$\sqrt{(2^2 + 15^2}$	15.13
B	(2, 55)	$\sqrt{(1^2 + 10^2}$	10.05
C	(3, 70)	$\sqrt{(0^2 + 5^2}$	5.00
D	(4, 80)	$\sqrt{(1^2 + 15^2}$	15.03
E	(5, 90)	$\sqrt{(2^2 + 25^2}$	25.08

4 Step 2: Sort the Distances

Sort the distances in ascending order (smallest first).

ID	Distance	Class
C	5.00	Pass (1)
B	10.05	Fail (0)
D	15.03	Pass (1)
A	15.13	Fail (0)
E	25.08	Pass (1)

5 Step 3: Select K Neighbors

We are using K = 3, so we take the 3 closest neighbors:

ID	Distance	Class
C	5.00	Pass (1)
B	10.05	Fail (0)
D	15.03	Pass (1)

These are the 3 nearest neighbors of the new point (3, 65).

6 Step 4: Make the Prediction

Use majority voting for classification.

- Among the 3 neighbors:
  - Pass (1) → 2 points (C and D)
  - Fail (0) → 1 point (B)

Prediction: Pass (1)

So, the new student is predicted to Pass.

For classification, we choose the class that appears most frequently among the K nearest neighbors.

7 Example (Regression)

Now let's see a simple regression example. Suppose we have house data:

ID	Size (m^2)	Price (USD)
A	50	100
B	60	120
C	70	160
D	80	200
E	90	240

Predict the price for a house with size = 75 m^2 using K = 2.

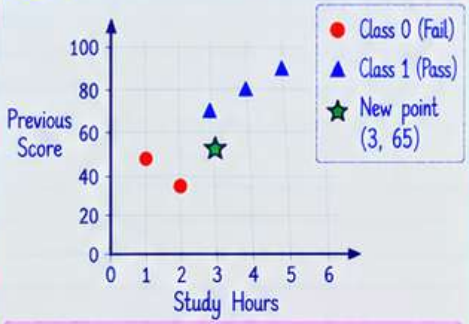
8 Steps for Regression

1. Calculate the distance between 75 and each training size.
2. Sort and select the 2 nearest neighbors.
3. Predict the price using the average of the neighbors' prices.

ID	Size	Distance	Price
A	50	25	100
B	60	15	120
C	70	5	160
D	80	5	200
E	90	15	240

The 2 nearest neighbors are C (70) and D (80).
Predicted price = (160 + 200) / 2 = 180 USD.

9 Visualization (Classification)



The new point is closer to Pass (1) points, so it is predicted as Pass.

10 Key Points to Remember

1. KNN is a non-parametric, instance-based learning algorithm.
2. It does not build an explicit model.
3. It uses distance to find similar data points.
4. K is a user-defined parameter.
5. For classification → use majority voting.
6. For regression → use the average of the neighbors.
7. Smaller distance means more similar points.
8. The choice of K can affect the prediction result.
9. It works well when similar data points belong to the same class or have similar output values.

11 Advantages and Disadvantages

Advantages	Disadvantages
✓ Simple and easy to understand.	✗ Can be slow for large datasets (must calculate distance to all points).
✓ No training phase.	✗ Sensitive to the choice of K.
✓ Works for both classification and regression.	✗ Affected by irrelevant features.
✓ Can handle multi-class problems.	✗ Does not work well with high-dimensional data (curse of dimensionality).
✓ Effective when similar data points are close.	

"KNN makes predictions by looking at the closest neighbors – simple but powerful!" ☺

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Lecture 04 – Step 3C

K-Nearest Neighbors (KNN) (Part 3)

Year 02
Semester 01
Date : .....

Choosing the Best K, Classification vs Regression and Important Points

"Better K
Better Predictions!" ☺

1 Classification vs Regression

- KNN can be used for both classification and regression problems.
- The prediction method is different for each case.

KNN Classification	KNN Regression
• Output is a category (class).	• Output is a continuous numeric value.
• Use majority voting (most frequent class).	• Use the average of the K nearest neighbors.
• Example: Pass / Fail, Yes / No, Class A / B.	• Example: house price, weight, temperature.

2 How to Choose the Best K?

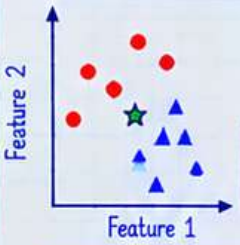
- K is a hyperparameter (chosen by the user).
- There is no single best value for K.
- A small K can be sensitive to noise (outliers).
- A large K can make the decision boundary too smooth and may ignore local patterns.
- The best K can be found using techniques like cross-validation.

Common choices:

- Use odd values for K in classification (e.g. 1, 3, 5, 7, ...).
- Try multiple K values and choose the one with the best performance (lowest error).

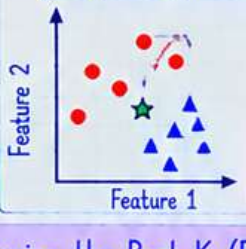
3 Effect of Different K Values

Small K (e.g. K = 1)



Very sensitive to noise (outliers).

Large K (e.g. K = 7)



Decision boundary becomes smoother (may ignore local patterns).

4 Example – Choosing the Best K (Classification)

We test different K values using cross-validation and get the following accuracy:

K	Accuracy
1	0.78
3	0.82
5	0.85
7	0.83
9	0.80

The best K is 5 because it gives the highest accuracy (0.85).

5 Example – Choosing the Best K (Regression)

- We test different K values and get the following Mean Squared Error (MSE):

K	MSE
1	120
3	85
5	60
7	65
9	80

The best K is 5 because it gives the lowest MSE (60).

6 Advantages and Disadvantages of KNN

Advantages	Disadvantages
✓ Simple to understand and implement.	✗ Computationally expensive for large datasets (need to calculate distances).
✓ Works for both classification and regression.	✗ Sensitive to the choice of K.
✓ No training phase (lazy learning).	✗ Affected by irrelevant features (need feature scaling).
✓ Can handle multi-class problems.	✗ Does not work well with high-dimensional data (curse of dimensionality).
✓ Works well when similar data points are close.	

7 Tips for Better Performance

1. Use cross-validation to find the best K.
2. Try odd values for K in classification.
3. Normalize or standardize the features (so all features have the same scale).
4. Remove irrelevant features.
5. Use an appropriate distance measure (Euclidean is common).
6. Consider the size of the dataset (KNN can be slow for very large datasets).

8 Key Points to Remember

1. KNN is a non-parametric, instance-based learning algorithm.
2. Used for both classification and regression.
3. Prediction is based on the K nearest neighbors.
4. For classification → use majority voting.
5. For regression → use the average of the neighbors.
6. K is a user-defined parameter (hyperparameter).
7. No single best K → use cross-validation.
8. Sensitive to feature scale and irrelevant features.
9. Works well when similar data points belong to the same class or have similar output values.

9 Summary

- KNN is simple but powerful.
- Choose the right K for better performance.
- Use majority voting for classification and average for regression.
- Use cross-validation to find the best K.
- Feature scaling and removing irrelevant features can improve results.
- Be careful with large datasets and high-dimensional data.

"KNN doesn't build a model,
it simply looks at the closest neighbors." ☺

"Right K, Right Neighbors, Better Predictions!" ☺

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Lecture 04 – Step 3D

K-Nearest Neighbors (KNN) – Part 4

Final Revision + Complete Example + Key Points

Year 02
Semester 01
Date : .....

"Practice Makes Perfect!" ☺

1 Complete KNN Process (Revision)

1. Choose the value of K.
2. Calculate the distance between the new point and all training points.
3. Sort the distances (smallest first).
4. Select the K nearest neighbors.
5. Make the prediction.

KNN is a lazy learning algorithm because it does not build a model, it just stores the training data and makes predictions when needed.

2 Complete Example (Classification)

Suppose we have the following training data with two features:

ID	x	y	Class
A	1	1	A
B	2	1	A
C	4	2	B
D	5	4	B
E	3	5	B

A new point is given: (2, 3)
Use K = 3 to predict the class.

Step 1: Calculate distances (Euclidean)

ID	x	y	Class	Distance
A	1	1	A	$\sqrt{5}$ = 2.24
B	2	1	A	$\sqrt{4}$ = 2.00
C	4	2	B	$\sqrt{5}$ = 2.24
D	5	4	B	$\sqrt{13}$ = 3.61
E	3	5	B	$\sqrt{5}$ = 2.24

3 Continue Example (Classification)

Step 2: Sort the distances

ID	Distance	Class
B	2.00	A
A	2.24	A
C	2.24	B
E	2.24	B
D	3.61	B

Step 3: Select K = 3 nearest neighbors

ID	Distance	Class
B	2.00	A
A	2.24	A
C	2.24	B

Step 4: Make the prediction (majority voting)

Class A → 2 points
Class B → 1 point

Predicted Class = A

4 Example (Regression)

Suppose we have the following training data (1 feature):

ID	Size (m²)	Price (USD)
A	50	100
B	60	120
C	70	160
D	80	200
E	90	240

A new house has size = 75 m².
Use K = 2 to predict the price.

Step 1: Calculate distances ($|x_2 - x_1|$)

ID	Size	Distance	Price
A	50	25	100
B	60	15	120
C	70	5	160
D	80	5	200
E	90	15	240

Step 2: Select K = 2 nearest neighbors

ID	Distance	Price
C	5	160
D	5	200

Step 3: Predict using average

(160 + 200) / 2 = 180

Predicted Price = 180 USD

5 Classification vs Regression

	KNN Classification	KNN Regression
Output	Category (class)	Continuous numeric value
Prediction method	Most frequent class (majority voting)	Average of the K nearest neighbors
Example	Pass / Fail, Yes / No, Class A / B	House price, Weight, Temperature

6 Choosing the Best K

- K is a hyperparameter (chosen by the user).
- There is no single best value for K.
- A small K can be sensitive to noise (outliers).
- A large K can make the decision boundary too smooth and may ignore local patterns.
- The best K can be found using techniques like cross-validation (hyperparameter optimization).

Common choice: Use odd values for K in classification (e.g. 3, 5, 7) to avoid ties.

7 Advantages and Disadvantages

Advantages	Disadvantages
✅ Simple and easy to understand.	❌ Computationally expensive for large datasets (need to calculate distances).
✅ No training phase.	❌ Sensitive to the choice of K.
✅ Works for both classification and regression.	❌ Affected by irrelevant features (need feature scaling).
✅ Can handle multi-class problems.	❌ Does not work well with high-dimensional data (curse of dimensionality).
✅ Works well when similar data points are close.	

8 Key Points to Remember

1. KNN is a non-parametric, instance-based learning algorithm.
2. It predicts based on the K nearest neighbors.
3. Use Euclidean distance to find similar data points.
4. For classification → use majority voting.
5. For regression → use the average of the neighbors.
6. K is a user-defined parameter.
7. No training phase (lazy learning).
8. The best K can be found using cross-validation.

9 Final Summary

- KNN is simple but powerful.
- It works for both classification and regression.
- It just looks at the closest neighbors.
- Choose the right K for better performance.
- Use feature scaling for better results.
- Be careful with large datasets and high-dimensional data.

Small Steps Lead to Big Results ! ☺

"KNN keeps it simple – learn from the neighbors!" ☺

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Lecture 04 – Step 4A

Decision Trees (Part 1)

Introduction, How it Works and CART

Year 02
Semester 01
Date : .....

"Simple Questions Lead to Big Decisions!" ☺

1 What is a Decision Tree?

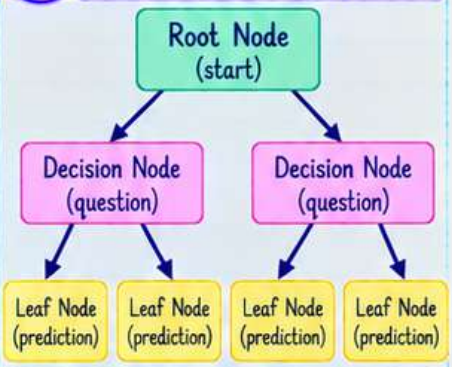
- A decision tree is a supervised learning algorithm used for classification and regression.
- It makes predictions by asking a series of simple questions about the input features.
- The model is represented as a tree structure with nodes, branches and leaves.
- It is easy to understand and can be visualized.

Main idea: "Split the data into smaller and more similar groups using simple rules."

2 When do we use Decision Trees?

- To solve classification problems (e.g. Yes / No, Class A / B).
- To solve regression problems (e.g. predict a numeric value).
- When we want a model that is easy to understand and explain.
- When the relationship between features and output is non-linear.
- It can handle both numerical and categorical data.

3 Tree Structure (Components)



Root Node: entire dataset
Decision Node: test a feature (split the data)
Leaf Node: final prediction (class or value)

4 How a Decision Tree Works?

1. Start with the entire training dataset at the root node.
2. Choose the best feature to split the data.
3. Split the data into smaller groups based on a condition (e.g. Age ≤ 30).
4. Repeat the process for each group (create more nodes).
5. Continue until a stopping condition is met (e.g. maximum depth or all data in a node belong to the same class).
6. The final leaf nodes give the predicted class or value.

Each split tries to make the groups as pure (similar) as possible.

5 Example (Classification)

Suppose we want to predict whether a person will play tennis based on weather conditions.

No.	Outlook	Temperature	Play Tennis
1	Sunny	Hot	No
2	Sunny	Mild	No
3	Overcast	Hot	Yes
4	Rainy	Mild	Yes
5	Rainy	Cool	Yes
6	Sunny	Cool	No

A decision tree can learn rules like:

If Outlook = Overcast → Play = Yes
If Outlook = Sunny → Play = No
If Outlook = Rainy → Play = Yes

6 Decision Tree for the Example



This tree gives a simple and easy-to-understand rule set for prediction.

7 Classification and Regression Trees (CART)

- CART (Classification and Regression Trees) is a popular method used to build decision trees.
- It can be used for both classification and regression.
- For classification, CART uses a measure like Gini Index to find the best split.
- For regression, CART uses a measure like Mean Squared Error (MSE) to find the best split.
- CART always creates binary splits (each node splits into two branches).
- It keeps splitting the data until a stopping condition is met.
- CART is simple and effective, and widely used in practice.

CART = A decision tree algorithm that works for both classification (Gini Index) and regression (MSE).

8 Key Points to Remember

1. A decision tree makes predictions using a tree structure.
2. It can be used for both classification and regression.
3. The tree consists of root node, decision nodes and leaf nodes.
4. Each internal node tests a feature and splits the data.
5. Leaf nodes give the final prediction.
6. CART is a popular decision tree algorithm.
7. CART uses Gini Index for classification and MSE for regression.
8. CART creates binary splits (two branches).
9. It is easy to understand and can handle both numerical and categorical data.
10. Decision trees can be visualized.
11. They work well for non-linear relationships.
12. Stopping conditions are used to prevent the tree from growing too large.

"A decision tree turns complex problems into simple questions." ☺

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Lecture 04 – Step 4B

Decision Trees (Part 2)

Gini Index, Purity Calculation and Best Split (CART)

Year 02
Semester 01
Date : .....

"Better Splits Create Purer Groups!" ☺

1 What is Gini Index?

- Gini Index is a measure of impurity.
- It is used in CART (Classification and Regression Trees) for classification.
- A lower Gini value means the node is more pure (more similar).
- Gini Index is 0 when all data in the node belong to the same class.
- Gini Index is highest when the classes are evenly mixed.

Goal: Choose the split that gives the lowest Gini Index (most pure groups).

2 Formula for Gini Index

For a node with k classes:

$$Gini = 1 - \sum_{i=1}^{k} p_i^2$$

where,

- k – number of classes
- p_i – proportion of data points in class i

For binary classification (2 classes):

$$Gini = 1 - (p^2 + q^2)$$

where p and q are the proportions of the two classes.

3 Gini Values – Examples

Pure node (all same class)

- Class A = 10, Class B = 0
- Gini = $1 - (1^2 + 0^2) = 0$
- Gini = 0 (completely pure)

Half and half (evenly mixed)

- Class A = 5, Class B = 5
- Gini = $1 - (0.5^2 + 0.5^2)$
  $= 1 - (0.25 + 0.25) = 0.5$
- Gini = 0.5 (most impure for 2 classes)

Mostly one class

- Class A = 8, Class B = 2
- Gini = $1 - (0.8^2 + 0.2^2)$
  $= 1 - (0.64 + 0.04) = 0.32$
- Gini = 0.32 (more pure than 0.5)

4 Example – Calculate Gini Index

A node has the following data:

Class	Count	Proportion (p)
A	8	8/10 = 0.8
B	2	2/10 = 0.2
Total	10	1.0

$Gini = 1 - (0.8^2 + 0.2^2)$
$= 1 - (0.64 + 0.04)$
$= 1 - 0.68$
$= 0.32$

Gini Index = 0.32

5 Splitting a Node

- We try different features and threshold values to split the data.
- For each possible split, calculate the Gini Index of the resulting child nodes.
- Choose the split with the lowest weighted Gini Index.

Weighted Gini Index:

$$Gini_{split} = \frac{n_{left}}{n} \times Gini_{left} + \frac{n_{right}}{n} \times Gini_{right}$$

where,

- n_{left} – number of data points in left node
- n_{right} – number of data points in right node
- n – total number of data points

6 Example – Find the Best Split

Suppose we have the following data for a feature (Age) and class (Play = Yes/No):

ID	Age	Play
1	10	No
2	12	No
3	20	Yes
4	22	Yes
5	25	Yes
6	30	Yes

Try a split at Age ≤ 18 and Age > 18.

- Left node (Age ≤ 18): 2 No, 0 Yes → Gini = 0
- Right node (Age > 18): 0 No, 4 Yes → Gini = 0
- Weighted Gini = (2/6 x 0) + (4/6 x 0) = 0

This is a perfect split! (Gini = 0)

7 Another Example (More Realistic)

Use the following data:

ID	Age	Play
1	10	No
2	14	No
3	16	Yes
4	20	Yes
5	22	Yes
6	24	No

Try split at Age ≤ 16 and Age > 16.

- Left node (Age ≤ 16): 2 No, 1 Yes
  $Gini_{left} = 1 - \left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = 0.44$
- Right node (Age > 16): 1 No, 2 Yes
  $Gini_{right} = 1 - \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = 0.44$
- Weighted Gini = (3/6 x 0.44) + (3/6 x 0.44)
  = 0.44

Weighted Gini = 0.44

8 Key Points to Remember

1. Gini Index measures impurity.
2. Lower Gini value means purer node.
3. $Gini = 1 - \sum_{i=1}^{k} p_i^2$.
4. CART uses Gini Index for classification.
5. Try different features and thresholds to find the best split.
6. Use weighted Gini Index to compare splits.
7. Choose the split with the lowest weighted Gini.
8. Gini is 0 for a pure node.
9. Gini is highest (0.5) when classes are evenly mixed (for 2 classes).
10. The process continues recursively until a stopping condition is met.

9 Summary

- Gini Index helps to find the best split in a decision tree.
- CART uses Gini Index.
- We calculate the Gini Index for each possible split and choose the one with the lowest weighted Gini.
- The goal is to create pure (homogeneous) groups.
- This process is repeated recursively to grow the tree.

"Lower Gini,
Better Split,
Purer Groups!" ☺

"A good split today, a smarter tree tomorrow!" ☺

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Lecture 04 – Step 4C

Decision Trees (Part 3)

Year 02
Semester 01
Date : .....

Example: Choosing the Best Split using Gini Index

"Use Gini to find the best feature!" ☺

1 Example Dataset (Buys_insurance)

We use the following dataset from the lecture:

ID	Age	Income	Student	Buys_insurance
1	<=30	High	No	No
2	<=30	High	No	No
3	31-40	High	No	Yes
4	>40	Medium	No	Yes
5	>40	Low	Yes	Yes
6	>40	Low	Yes	No
7	31-40	Low	Yes	Yes
8	<=30	Medium	No	No
9	<=30	Low	Yes	Yes
10	>40	Medium	Yes	Yes

2 Gini Index at the Root Node

At the root node, we have all 10 records.
Class distribution:

- Yes = 5
- No = 5

$$Gini = 1 - (p_{yes}^2 + p_{no}^2)$$

$$P_{yes} = \frac{5}{10} = 0.5$$

$$P_{no} = \frac{5}{10} = 0.5$$

$$Gini = 1 - (0.5^2 + 0.5^2)$$
$$= 1 - (0.25 + 0.25)$$
$$= 1 - 0.5$$
$$= 0.5$$

Root node Gini = 0.5 (which is impure)

3 Split on Age

(a) Age <= 30

Buys_insurance	Count
Yes	1
No	3
Total	4

$Gini = 1 - (0.25^2 + 0.75^2)$
$= 1 - (0.0625 + 0.5625)$
$= 0.375$

(b) Age 31 – 40

Buys_insurance	Count
Yes	2
No	0
Total	2

$Gini = 1 - (1^2 + 0^2)$
$= 0$ (pure node)

(c) Age > 40

Buys_insurance	Count
Yes	3
No	1
Total	4

$Gini = 1 - (0.75^2 + 0.25^2)$
$= 1 - (0.5625 + 0.0625)$
$= 0.375$

Weighted Gini (Age) = (4/10 × 0.375) + (2/10 × 0) + (4/10 × 0.375)
= 0.15 + 0 + 0.15 = 0.30

4 Split on Income

(a) Income = High

Buys_insurance	Count
Yes	1
No	2
Total	3

$Gini = 1 - (1/3)^2 - (2/3)^2$
$= 1 - (0.11 + 0.44)$
$= 0.44$

(b) Income = Medium

Buys_insurance	Count
Yes	2
No	1
Total	3

$Gini = 1 - (2/3)^2 - (1/3)^2$
$= 1 - (0.44 + 0.11)$
$= 0.44$

(c) Income = Low

Buys_insurance	Count
Yes	2
No	1
Total	3

$Gini = 1 - (2/3)^2 - (1/3)^2$
$= 1 - (0.44 + 0.11)$
$= 0.44$

Weighted Gini (Income) = (3/10 × 0.44) + (3/10 × 0.44) + (3/10 × 0.44)
= 0.132 + 0.132 + 0.132 = 0.396 ≅ 0.40

5 Split on Student

(a) Student = Yes

Buys_insurance	Count
Yes	3
No	1
Total	4

$Gini = 1 - (0.75^2 + 0.25^2)$
$= 0.375$

(b) Student = No

Buys_insurance	Count
Yes	2
No	3
Total	5

$Gini = 1 - (0.4^2 + 0.6^2)$
$= 1 - (0.16 + 0.36)$
$= 0.48$

Weighted Gini (Student) = (4/10 × 0.375) + (5/10 × 0.48)
= 0.15 + 0.24 = 0.39

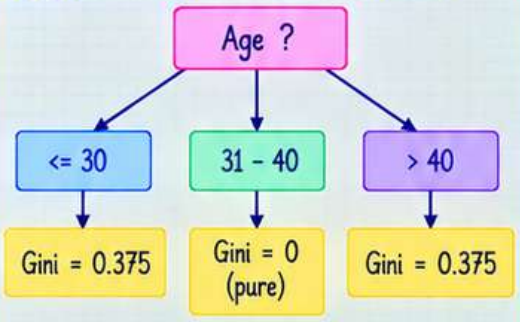
6 Compare and Choose the Best Split

Feature	Weighted Gini
Age	**0.30**
Income	0.40
Student	0.39

The lowest weighted Gini is for **Age (0.30)**, so we choose *Age* as the best feature to split.

Rule:
Choose the feature with the lowest weighted Gini Index.

7 Resulting Tree (First Split)



8 Key Points to Remember

1. Gini Index measures impurity.
2. Gini = 0 means a pure node.
3. Lower Gini is better.
4. Calculate Gini for each possible split.
5. Use weighted Gini to compare features.
6. Choose the feature with the lowest weighted Gini.
7. CART uses Gini Index.
8. The splitting process continues recursively until a stopping condition is met.

9 Summary

- ✅ We calculated Gini Index for the root node.
- ✅ We calculated Gini for each value of a feature.
- ✅ We used weighted Gini to compare features.
- ✅ Age gave the lowest weighted Gini (0.30).
- ✅ So, Age is the best split.
- ✅ The tree is built recursively.

"A good decision tree chooses the split that creates purer groups!" ☺

"Lower Gini → Purer Groups → Better Decisions!" ☺

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Lecture 04 – Step 4E

Decision Trees (Part 4)

Final Revision + Complete Tree + Exam Practice

Year 02
Semester 01
Date : .....

"Practice Today
Better Tomorrow!" ☺

1 Final Dataset (Buys_insurance)

ID	Age	Gender	Income	Credit_Score	Buys_insurance
1	20	Male	High	Excellent	Yes
2	25	Female	Medium	Excellent	Yes
3	30	Male	High	Fair	No
4	35	Female	Low	Fair	No
5	40	Male	Medium	Poor	No
6	45	Female	Low	Excellent	Yes

Target variable: Buys_insurance (Yes / No)
Features: Age (numeric), Gender (categorical), Income (categorical), Credit_Score (categorical)

2 Credit Score – Gini Calculation

Credit_Score	Yes	No	Gini
Excellent	3	0	0
Fair	0	2	0
Poor	0	1	0

Weighted Gini (Credit_Score):

$$\left(\frac{3}{6} \times 0\right) + \left(\frac{2}{6} \times 0\right) + \left(\frac{1}{6} \times 0\right) = 0$$

Credit Score gives Gini = 0 (completely pure)!"

3 Age – Find Best Threshold

Sorted ages: 20, 25, 30, 35, 40, 45

Possible thresholds (average of consecutive):

(20 + 25) / 2 = 22.5
(25 + 30) / 2 = 27.5
(30 + 35) / 2 = 32.5
(35 + 40) / 2 = 37.5
(40 + 45) / 2 = 42.5

Threshold (Age)	Weighted Gini
22.5	0.40
27.5	0.50
32.5	0.444
37.5	0.25
42.5	0.40

Best threshold = 37.5 (with lowest Gini = 0.25)

4 Gender – Gini Calculation

Gender	Yes	No	Gini
Male	2	1	0.444
Female	1	2	0.444

Weighted Gini (Gender):

$$\left(\frac{3}{6} \times 0.444\right) + \left(\frac{3}{6} \times 0.444\right) = 0.444$$

Gender Gini = 0.444

5 Income – Gini Calculation

Income	Yes	No	Gini
High	2	1	0.444
Medium	1	0	0
Low	0	2	0

Weighted Gini (Income):

$$\left(\frac{3}{6} \times 0.444\right) + \left(\frac{1}{6} \times 0\right) + \left(\frac{2}{6} \times 0\right) = 0.222$$

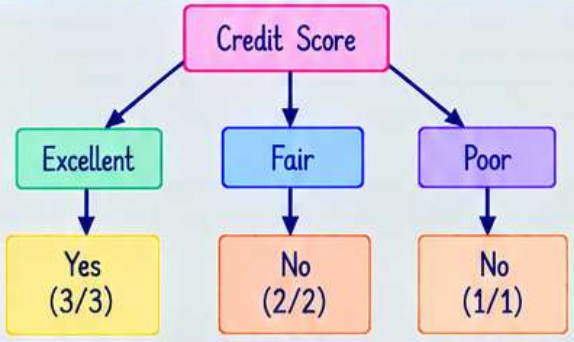
Income Gini = 0.222

6 Compare All Features

Feature	Weighted Gini
Gender	0.444
Income	0.222
Age (best threshold 37.5)	0.25
Credit Score	0

Best feature = Credit Score (since it has the lowest Gini = 0)

7 Resulting Decision Tree (Root = Credit Score)



The tree stops here because all leaf nodes are pure (Gini = 0). No further splitting is needed.

8 Key Points to Remember

1. Decision Trees are non-parametric supervised learning algorithms.
2. They can be used for both classification and regression.
3. The tree consists of a root node, decision nodes and leaf nodes.
4. Gini Index measures impurity.
5. Lower Gini indicates a purer node.
6. For categorical features, calculate Gini for each group and use weighted Gini.
7. For numerical features, sort the values and use the average of consecutive values as thresholds.
8. Choose the feature with the lowest weighted Gini for splitting.
9. CART (Classification and Regression Trees) uses Gini Index.
10. The process continues recursively until a stopping condition is met.

9 Possible Exam Questions

1. Define a decision tree and explain its components.
2. What is Gini Index? Give the formula.
3. Calculate the Gini Index for a given dataset.
4. Given a dataset, find the best split using Gini Index.
5. Explain how to find threshold values for a numerical feature.
6. Compare features and identify the best root node.
7. Draw the decision tree for the given example.
8. What is CART? Which measure does it use?

10 Final Summary

- ✅ Decision Trees are easy to understand and interpret.
- ✅ Gini Index helps to find the best split.
- ✅ We calculate Gini for each feature and choose the lowest.
- ✅ For numerical features, we test different threshold values.
- ✅ In the example, Credit Score gave the lowest Gini (0), so it was chosen as the root node.
- ✅ The tree is simple and stops when nodes are pure.
- ✅ CART uses Gini Index.

Simple Steps Powerful Results! ☺

Better Splits → Purer Groups → Better Predictions!" ☺

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Step 5 - Support Vector Machine (SVM)

Short Note

Year 02
Semester 01

"Find the best boundary, for better predictions!" ☺

1 What is SVM?

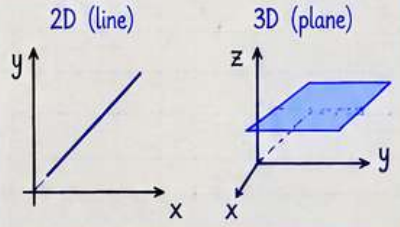
- SVM (Support Vector Machine) is a supervised learning algorithm.
- It finds a decision boundary (hyperplane) that separates different classes.
- It tries to maximize the margin (distance between boundary and closest data points).
- Can be used for classification and regression (mainly classification).

Main idea: "Find the optimal separating hyperplane with maximum margin."

2 Hyperplane

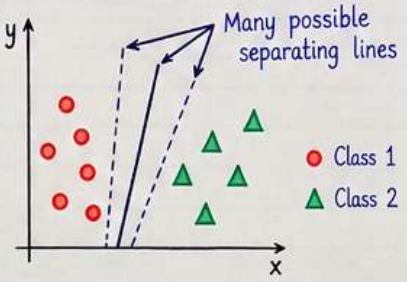
- A hyperplane is the decision boundary that separates classes.
- In 2 dimensions → line
- In 3 dimensions → plane
- In n dimensions → (n-1) dimensional hyperplane.

Examples:



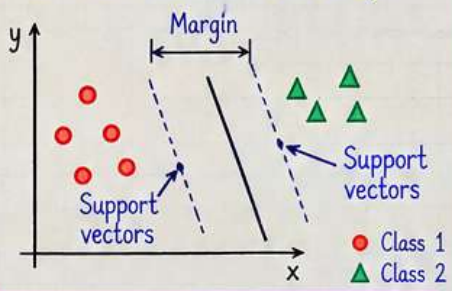
3 Separating Hyperplane

- A hyperplane that separates different classes.
- There can be many possible separating hyperplanes.
- SVM chooses the **best** one (with maximum margin).



4 Margin

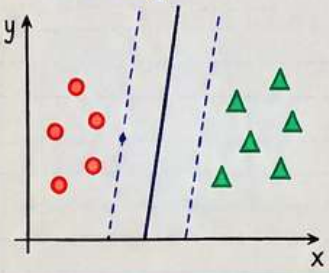
- Margin is the perpendicular distance between the hyperplane and the closest data points.
- The points closest to the hyperplane are called support vectors.
- SVM tries to maximize this margin.



Larger margin → better generalization (lower chance of overfitting).

5 Linear Separable Case

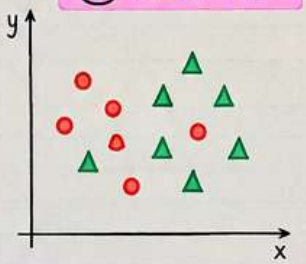
- Classes can be perfectly separated by a straight hyperplane.
- There may be many separating hyperplanes.
- SVM chooses the one with the maximum margin.



Goal: Correct classification + Maximum margin

6 Linear Non-Separable Case

- In real datasets, data may not be perfectly separable.
- A single straight hyperplane cannot classify all points correctly.
- SVM handles this using:
  1. Soft Margin
  2. Kernel Trick



Data is mixed → not linearly separable.

7 Soft Margin

- Allows some points to be misclassified.
- SVM tries to balance between:
  - Maximizing the margin
  - Minimizing misclassification.
- Two types of misclassified points:
  1. Point on the wrong side but within the margin.
  2. Point on the wrong side and also beyond the margin.

Useful when data is not perfectly separable.

8 Kernel Trick

- Transforms existing features into new higher-dimensional features.
- Helps SVM find a non-linear decision boundary.
- Common kernels:
  - Linear Kernel
  - Polynomial Kernel
  - RBF (Radial Basis Function) Kernel
  - Sigmoid Kernel

Kernel Trick allows SVM to handle complex non-linear data.

9 Key Points to Remember

1. SVM is a supervised learning algorithm.
2. Uses a hyperplane to separate classes.
3. Margin = distance to closest points.
4. Support vectors are the closest points.
5. Choose the hyperplane with maximum margin.
6. Works for both separable and non-separable data.
7. Soft margin allows some misclassification.
8. Kernel trick helps to handle non-linear data.

10 Short Exam Answer (Definition)

"SVM is a machine learning algorithm that finds an optimal hyperplane to separate different classes by maximizing the margin."

SVM → Maximum Margin → Better Generalization → Better Predictions! ☺

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Lecture 04 - Complete Short Note

Summary of All Algorithms

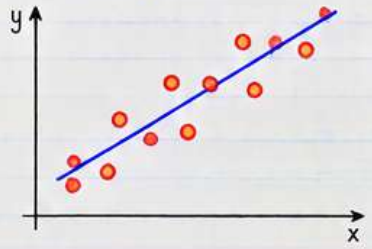
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Semester 01
Date : .....

"Small Notes
Big Results!" ☺

1 Linear Regression

- Used for predicting continuous numerical values.
- Finds a linear relationship between independent and dependent variables.
- **Model:** $y = b_0 + b_1 x$

y → predicted value
b_0 → intercept
b_1 → slope (coefficient)
x → independent variable



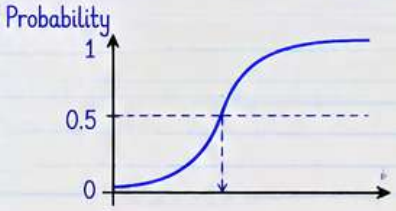
Goal: Minimize the error between predicted and actual values.

2 Logistic Regression

- Used for classification (mainly binary: 0 or 1).
- Predicts probability using sigmoid function.
- **Model:**

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$$

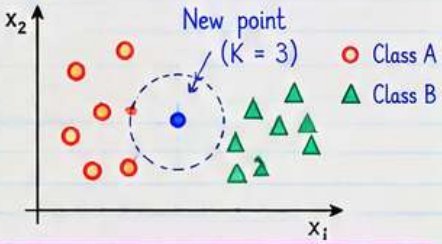
- **Output:** probability between 0 and 1.
- **Classify:**
  - $P \geq 0.5$ → Class 1 (e.g. Yes)
  - $P < 0.5$ → Class 0 (e.g. No)



Goal: Classify data based on probability.

3 K-Nearest Neighbors (KNN)

- A non-parametric, instance-based learning algorithm.
- Classifies a new data point based on the majority class of its K nearest neighbors.
- **Steps:**
  1. Choose value of K.
  2. Calculate distance to all points (e.g. Euclidean distance).
  3. Find K nearest neighbors.
  4. Choose the majority class.

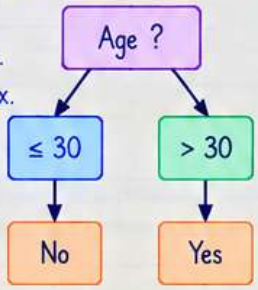


Goal: Classify based on nearest neighbors.

4 Decision Trees

- A tree-like model used for classification and regression.
- Data is split based on feature values.
- Common splitting measure: Gini Index.
- Gini formula (for k classes):

$$Gini = 1 - \sum_{i=1}^{k} p_i^2$$

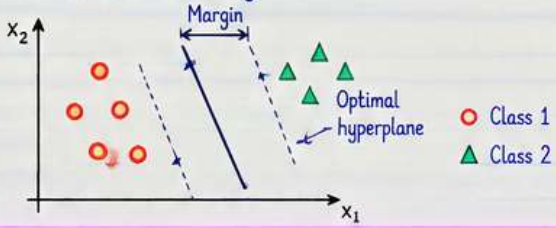


- Lower Gini → purer node.
- Choose the split with the lowest Gini Index.
- Process continues recursively to grow the tree.

Goal: Create pure groups by choosing the best splits (using lowest Gini Index).

5 Support Vector Machine (SVM)

- A supervised learning algorithm used for classification (and regression).
- Finds the optimal separating hyperplane between classes by maximizing the margin.
- Hyperplane: decision boundary (line in 2D, plane in 3D).
- Margin: distance between hyperplane and closest points (support vectors).
- Linear separable case: choose hyperplane with maximum margin.
- Non-separable case:
  - Soft margin → allow some misclassification.
  - Kernel trick → transform features to find a non-linear decision boundary.



Goal: Maximize the margin for better generalization.

Final Summary → Key Differences

Algorithm	Task	Output	Key Idea
Linear Regression	Regression	Continuous value	Fit a line (minimize error)
Logistic Regression	Classification	Probability (0 or 1)	Sigmoid function
KNN	Classification	Class label	Nearest neighbors
Decision Trees	Classification / Regression	Class label / value	Split data (Gini Index)
SVM	Classification	Class label	Optimal hyperplane (max margin)

"Learn Today
Apply Tomorrow
Build a Smarter You!"
☺

Same Concepts
Brighter Future! ☺